

JEE MAIN REVISION MASTER: TRIGONOMETRIC RATIOS & IDENTITIES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Compound, Multiple & Submultiple Angles	13 PYQs	Trigonometric Identities

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Compound, Multiple & Submultiple Angles	$\sin 18^\circ = \frac{\sqrt{5}-1}{4}$, $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$, $\tan(A+B+C) = \frac{S_1 - S_3}{1 - S_2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Compound, Multiple & Submultiple Angles (High Yield — 13 PYQs)

Core Principle: Core identities: $\sin 2\theta = 2\sin\theta\cos\theta = \frac{2\tan\theta}{1+\tan^2\theta}$, $\cos 2\theta = \cos^2\theta - \sin^2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}$. Standard angle values: $\sin 15^\circ = \cos 75^\circ = \frac{\sqrt{6}-\sqrt{2}}{4}$, $\cos 15^\circ = \sin 75^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}$, $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$.

■ **Caution / JEE Pitfall:** Watch for signs in different quadrants: ASTC rule (All, Sin, Tan, Cos). In quadrant II, only sine and cosecant are positive.

■ **Speed Trick:** If $A + B + C = 180^\circ$, then $\tan A + \tan B + \tan C = \tan A \tan B \tan C$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If α and β are acute angles such that $\alpha-\beta$ and $\alpha+\beta$ satisfy the equation $\tan^2\theta-4\tan\theta+1=0$, then..." [Ans: A]
- [2026 Shift] Q#2 (MCQ): "The maximum value of $\sin(x+(\pi/6))+\cos(x+(\pi/6))$ is attained at $x=...$ " [Ans: D]
- [2025 Shift] Q#3 (MCQ): "Which of the following is not correct?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If α and β are acute angles such that $\alpha - \beta$ and $\alpha + \beta$ satisfy the equation $\tan^2\theta - 4\tan\theta + 1 = 0$, then α and β are respectively,

- (A) $45^\circ, 30^\circ$ **[Correct]**
- (B) $30^\circ, 45^\circ$
- (C) $30^\circ, 60^\circ$
- (D) $60^\circ, 45^\circ$

Correct Answer: A

Solution Analysis: Put $x = \tan\theta$. The given equation is $\tan^2\theta - 4\tan\theta + 1 = 0$ So, $x^2 - 4x + 1 = 0$ Using the quadratic formula, $x = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$ Now, $\tan 15^\circ = 2 - \sqrt{3}$ and $\tan 75^\circ = 2 + \sqrt{3}$ So the two angles satisfying the equation are 15° and 75° Here the angles are $\alpha - \beta$ and $\alpha + \beta$. Since $\alpha + \beta$ is greater than $\alpha - \beta$, $\alpha - \beta = 15^\circ$ and $\alpha + \beta = 75^\circ$ Adding both equations, $2\alpha = 90^\circ$ $\alpha = 45^\circ$ Now, $\alpha - \beta = 15^\circ$